

INDIAN INSTITUTE OF TECHNOLOGY,
MADRAS

A mid-term report for the BDM capstone Project



Enhancing Operational Efficiency of a Soft Drink Wholesale Store through
Data Management

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EXECUTIVE SUMMARY:

This report presents an analysis of data obtained from SHIVA ENTERPRISES, a local wholesale softdrink store dealer located in Yusufpur, Ghazipur, Uttar Pradesh. The business is B2B as well as B2C and deals in the segment of drinkable items. This project aims to analyze sales of a soft drink wholesale store to identify the opportunities and area of improvement and try to provide solutions of it. The store faces a vital hurdle with immoderate inventory collection due to inaccurate demand forecasting and flawed inventory practices. This results in tied-up financial resources and impacts overall profitability. Furthermore, operational incompetency increases the problem, including order processing and inconsistent supply chain management. The analysis utilizes available data on daily sales and purchase records, profit and loss analysis for three months (July-September) of financial year 2023-24. We aim to propose pricing strategies aligned with pricing trends, and customer preferences, while maintaining a healthy profit margin. To overcome the hurdle of limited market dominance, we will conduct a market analysis to identify major preferences and demand. Moreover, the analysis highlights the necessity for more detailed data on sales categories and quantities to effectively tackle the remaining challenges. Gathering this data will provide insights for pricing optimization, exploring new sales channel and effectively managing costs. The strategies we propose, based on data analysis, have the potential to turn the SHIVA ENTERPRISES into a vibrant and efficient store within the operational area. This analysis provides the basis for making informed decisions and strategic planning. It equips Shiva Enterprises with actionable insights to drive growth, profitability, and sustainability in the competitive market.

PROOF OF ORIGINALITY OF THE DATA:

The data collected by me is primary and raw data, which means the data is collected by one-on-one interaction with the business owner. Since he keeps the data of sales and costs in notebooks and daily maintain it, so he took particular data from notebook and gave it to me. The data was in raw format and with limited schema, and it has been arranged wherever possible to excel and remaining analysis is done in excel and with python tools too.

A survey link was not required in this situation and hence it is not used.

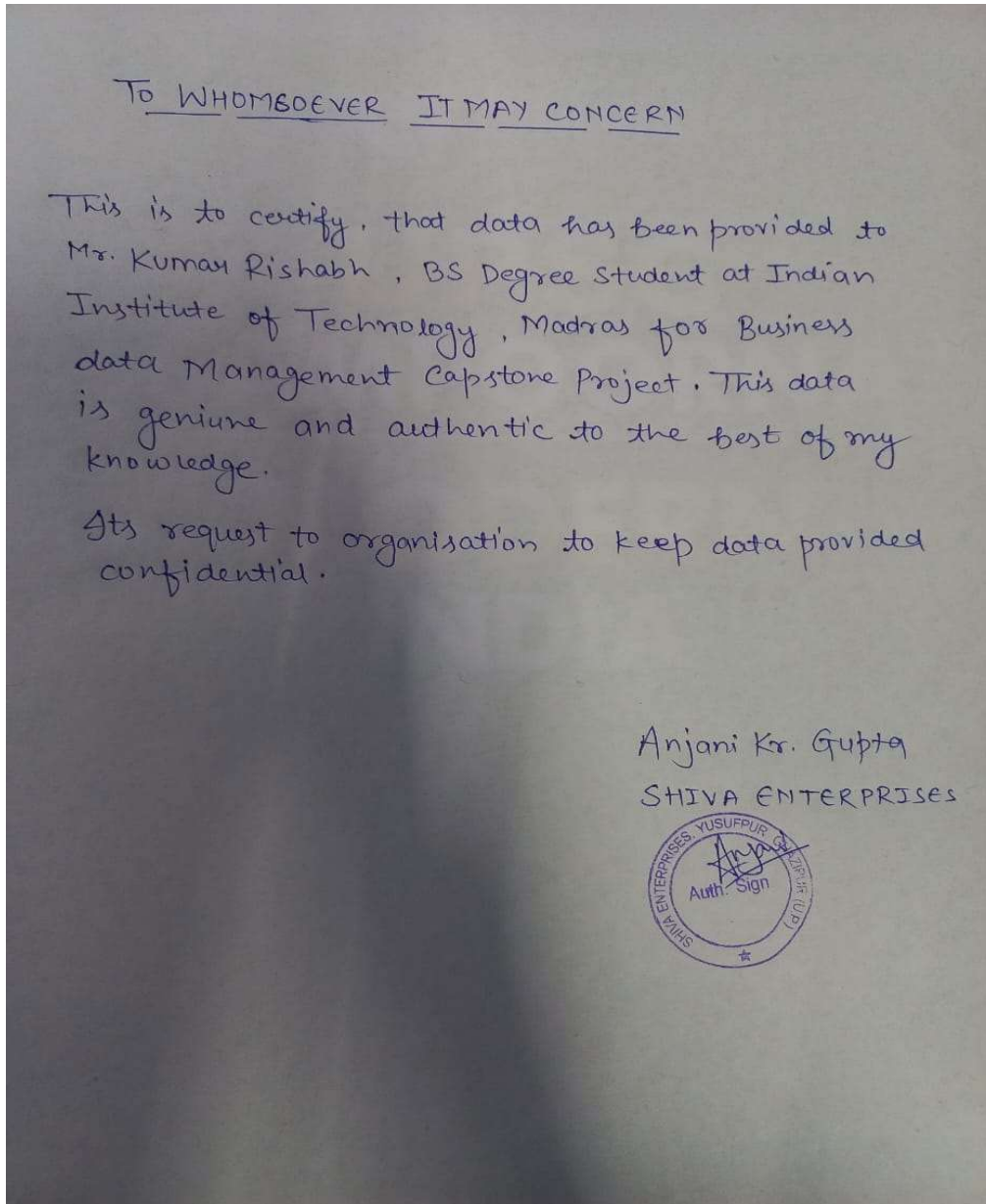
Recorded video with the Mr. Anjani Kumar Gupta (uploaded in drive link below):

https://drive.google.com/file/d/1WgEs07CTSJv6VgT6tWabGUsRcw_N0O2y/view?usp=sharing

In the video, Mr. Anjani Gupta talks about:

- i. The Store details, when it was started
- ii. The way sale happens in this wholesale store, problems faced by them.

Here is the letter from the business organization granting permission of using their data for the project.



Pictures showing the wholesale store and the products in store, along with me and Mr. Anjani Kumar Gupta.

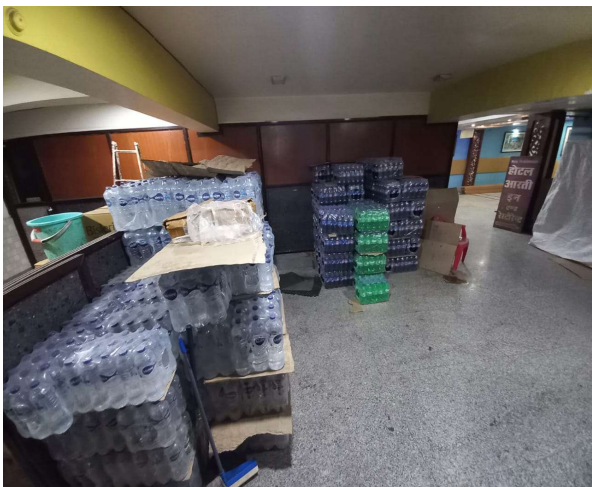


Fig: Store



Fig: myself in store



Fig: Owner

30

AUGUST

2023

WEDNESDAY

11 कार्यालय, मुंबई, ए. ई. ई.

4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

BIRLA CORPORATION LIMITED

AUGUST

THURSDAY

11 कार्यालय, मुंबई, ए. ई. ई.

2023

Amount

1000

METADATA:

Dataset Name: BDM Project Dataset

Link: <https://docs.google.com/spreadsheets/d/1aNtF1xGXQK7M7x7za4Vv1vkLdWa7eu8iLol7pma3Tzs/edit?usp=sharing>

Description: This dataset contains all transactional information of three months related to the SHIVA ENTERPRISES. Using Microsoft excel, the raw data was processed by adding all necessary columns. The schema provides information about the date of transaction, customer name, company name and name of product, package category, purchasing rate, selling rate, quantities of product sold, mode of payment and location of customer as well. This schema helps in analysis and decision making related to pricing and discount strategies, inventory management and market share.

Time Period: The dataset contains data of three months from July 1, 2023, to September 30, 2023 covering a range of transaction dates.

Schema:

	A	B	C	D	E	F	G	H	I	J
1	DATE	CUSTOMER NAME	COMPANY	PRODUCT	PACKAGE CATEGORY	PURCHASE RATE	SELLING RATE	QUANTITY(BOXES)	MODE OF PAYMENT	LOCATION
2	01-07-2023	AMARNATH	COCA-COLA	SPRITE	1.25L	640	690	3	UPI	NAVAPURA
3	01-07-2023	AMARNATH	COCA-COLA	SPRITE	2.25L	750	810	2	UPI	NAVAPURA
4	01-07-2023	AMARNATH	PARLE AGRO	BAILEY	250ML	120	156	12	UPI	NAVAPURA
5	01-07-2023	AMARNATH	PEPSICO	PEPSI	750ML	440	570	4	UPI	NAVAPURA
6	01-07-2023	AKRAM	COCA-COLA	SPRITE	750ML	510	570	3	UPI	BALAPUR
7	01-07-2023	AKRAM	COCA-COLA	THUBS UP	2L	660	815	2	UPI	BALAPUR
8	01-07-2023	SUNIL	COCA-COLA	SPRITE	750ML	510	570	3	UPI	SHERPUR

Columns:

1. **Date:** The date of each transaction.
2. **Customer Name:** Name of customer involved.
3. **Company:** Name of company whose product is being sold
4. **Product:** Name of particular product being sold.
5. **Package Category:** The net content volume of product of being sold.
6. **Purchase Rate:** The purchasing rate of product.
7. **Selling Rate:** The selling rate of product.
8. **Quantity:** The quantity of package sold.
9. **Mode of Payment:** The mode of payment for transaction.
10. **Location:** The location of the customer.

This metadata provides an overview of the structure of data included in the dataset provided in EXCEL workbook. The dataset can be used to analyse sales patterns, product

demand, customer preferences, profitability, and operational efficiency of the SHIVA ENTERPRISES.

DESCRIPTIVE STATISTICS:

It involves summarization and organisation of important aspects of dataset (such as mean, median, mode etc..) to provide a concise and meaningful representation of characteristics of the dataset. These statistics help to understand typical variability and distribution of data, by providing valuable insights for making business decision and planning.

	PURCHASE RATE	SELLING RATE	QUANTITY (BOXES)	TOTAL	DAILY PROFIT
count	887.000000	887.000000	887.000000	887.000000	887.000000
mean	485.631342	566.844419	3.670800	1655.458850	258.113867
std	265.317427	293.118283	3.666681	1329.240139	198.350573
min	120.000000	155.000000	1.000000	180.000000	40.000000
25%	360.000000	430.000000	2.000000	810.000000	120.000000
50%	480.000000	570.000000	3.000000	1630.000000	180.000000
75%	630.000000	760.000000	4.000000	2040.000000	390.000000
max	1260.000000	1440.000000	18.000000	8640.000000	1080.000000

Fig: Descriptive Stats

i. Purchase Rate

- Count: 887 entries
- Mean: The average purchase rate is approximately 485.63.
- Standard Deviation: The purchase rate has standard deviation of approximately 265.32.
- Minimum: The minimum purchase rate is 120.00
- 25th percentile: 25th percentile of purchase rate of dataset is approximately 360.00 or lower.
- 50th percentile: Median of purchase rate of dataset is approximately 480.00.
- 75th percentile: 75th percentile of purchase rate of dataset is 630.00.
- Maximum: The Maximum purchase rate is 1260.00.

ii. Selling Rate

- Count: 887 entries.
- Mean: The average selling rate is approximately 566.84
- Standard Deviation: The selling rate has standard deviation of approximately 293.11.
- Minimum: The minimum selling rate is 155.00
- 25th percentile: 25th percentile of selling rate of dataset is approximately 430.00 or lower.
- 50th percentile: Median of selling rate of dataset is approximately 570.00.
- 75th percentile: 75th percentile of selling rate of dataset is 760.00.
- Maximum: The Maximum selling rate is 1440.00.

iii. Quantity

- Count: 887 entries.
- Mean: The average of quantity is approximately 3.67
- Standard Deviation: The quantity has standard deviation of approximately 3.67.
- Minimum: The Minimum quantity is 1.00.
- 25th percentile: 25th percentile of quantity is approximately 2.00 or lower.
- 50th percentile: Median of quantity is approximately 3.00.
- 75th percentile: 75th percentile of quantity is 4.00.
- Maximum: The Maximum quantity is 18.00.

iv. Total

- Count: 887 entries.
- Mean: The average of total is approximately 1655.46
- Standard Deviation: The Total has standard deviation of approximately 1329.24.
- Minimum: The minimum of total is 180.00.
- 25th percentile: 25th percentile of Total is approximately 810.00 or lower.
- 50th percentile: Median of Total is approximately 1630.00.
- 75th percentile: 75th percentile of Total is 2040.00.
- Maximum: The Maximum of Total is 8640.00.

v. Daily Profit

- Count: 887 entries.
- Mean: The average of Daily profit is approximately 258.11
- Standard Deviation: The Daily profit has standard deviation of approximately 198.35.
- Minimum: The minimum of Daily profit is 40.00.
- 25th percentile: 25th percentile of Daily profit is approximately 120.00 or lower.
- 50th percentile: Median of Daily is approximately 180.00.
- 75th percentile: 75th percentile of Daily profit is 390.00.
- Maximum: The Maximum daily profit is 1080.00.

DETAILED EXPLANATION OF ANALYSIS PROCESS/METHOD:

During data analysis and arrangement phase, the raw data went through several transformation to gain key insights related to store's financial and other performances. One of the major transformations is addition of two new columns namely "Total" and "Daily Profit" to enhance the analytical capabilities of dataset. The "Total" column calculated the price of product by using its Selling Rate and Quantity of items sold. The "Daily Profit" column provides the information of profit generated by store on daily basis. It can be calculated by subtracting Purchase Rate from Selling Rate and multiplying by Quantity of product sold. This column helps to understand store's profitability trend, financial performance over time and making decisions on profit fluctuations. Furthermore, more columns namely "Day", "Month", "Company" were added, which helps in gaining the Day and Month wise information about customer engagement, demand and sales, profitability. "Company" will provide insight about parent company of products to analyze demand and sales of a product of particular company.

1	DATE	DAY	MONTH	CUSTOMER NAME	COMPANY	PRODUCT	CATEGORY	PACKAGE CATEGORY	PURCHASE RATE	SELLING RATE	QUANTITY(BOXES)	MODE OF PAYMENT	TOTAL	DAILY PROFIT	LOCATION
2	01-07-2023	Saturday	July	AMARNATH	COCA-COLA	SPRITE	SOFT DRINKS	1.25L	640	690	3	UPI	2070	150	NAVAPURA
3	01-07-2023	Saturday	July	AMARNATH	COCA-COLA	SPRITE	SOFT DRINKS	2.25L	750	810	2	UPI	1620	120	NAVAPURA
4	01-07-2023	Saturday	July	AMARNATH	PARLE AGRO	BAILEY	WATER	250ML	120	156	12	UPI	1872	432	NAVAPURA
5	01-07-2023	Saturday	July	AMARNATH	PEPSICO	PEPSI	SOFT DRINKS	750ML	440	570	4	UPI	2280	520	NAVAPURA
6	01-07-2023	Saturday	July	AKRAM	COCA-COLA	SPRITE	SOFT DRINKS	750ML	510	570	3	UPI	1710	180	BALAPUR
7	01-07-2023	Saturday	July	AKRAM	COCA-COLA	THUBS UP	SOFT DRINKS	2L	660	815	2	UPI	1630	310	BALAPUR

Fig: Processed Data

The analysis procedure involved the following steps, by using Python and Excel functions and its methodologies to extract meaningful findings from available data:

- 1. Data Preparation:** Arrangement of raw data provided in Excel, inspection of data for data type, inconsistency, to add columns to make data more informative. Cleaning and processing the data by handling missing values, correcting format issues, and ensuring data consistency.
- 2. Descriptive Statistics:** Loading the dataset into a data analysis tool such as Python with libraries like Pandas and NumPy or spreadsheet software. Calculate statistics summary for key variables. Visualise the variable distribution using density plots, histogram etc. to analyse their central tendency.
- 3. Sales and Revenue Analysis:** Calculate daily sales and revenue of data to identify overall performance by store in each product category. Analyse the trend in daily sales and revenue and plot graph to visualise changes. Line chart were generated to analyse trend of sales over the period of three months.

4. Profit and Demand Analysis: Calculate the profit margin on sales of each product by subtracting costs from net sales. Analyse the profitability of different product by calculating profit margin for each product. Identify the most profitable and least profitable product from data provided. For demand, identify the most and least sold products based on the quantity sold Analyse the distribution of sold products to understand demand patterns.

5. Visualisation, Insights and Recommendation: By creating charts such as line charts, bar plots, pie charts, scatter plots, we can represent various aspects of analysis. Visualise revenue, sales, demand over the time to identify gain or loss patterns.

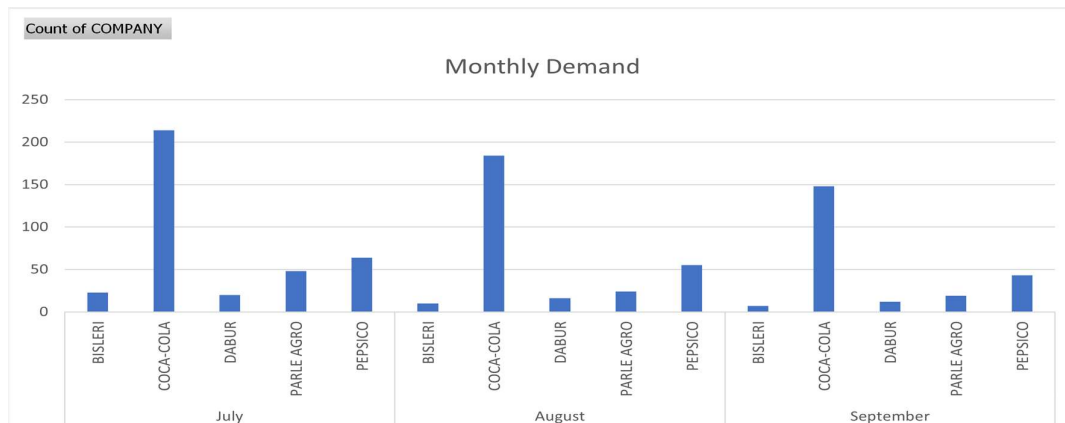
Give recommendations to improve stock management, marketing and pricing strategies, operational efficiencies based on analysis result.

By following this analysis method, meaningful data can be gathered from the provided dataset. The sales analysis provides overview of monthly trends, statistics offers quantitative measures of the provided data, profit analysis provide overview of profit margin on each product.

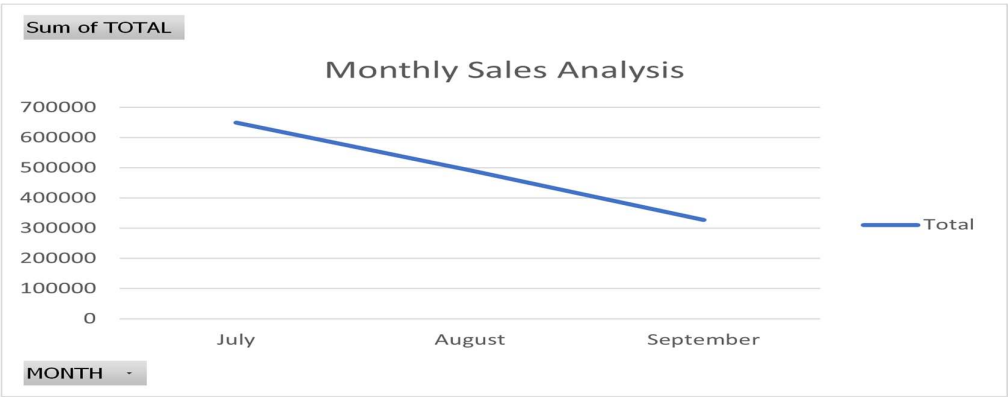
The findings can guide in inventory management, sales and demand forecasting to smoothing operations and drive business growth.

RESULTS AND FINDINGS:

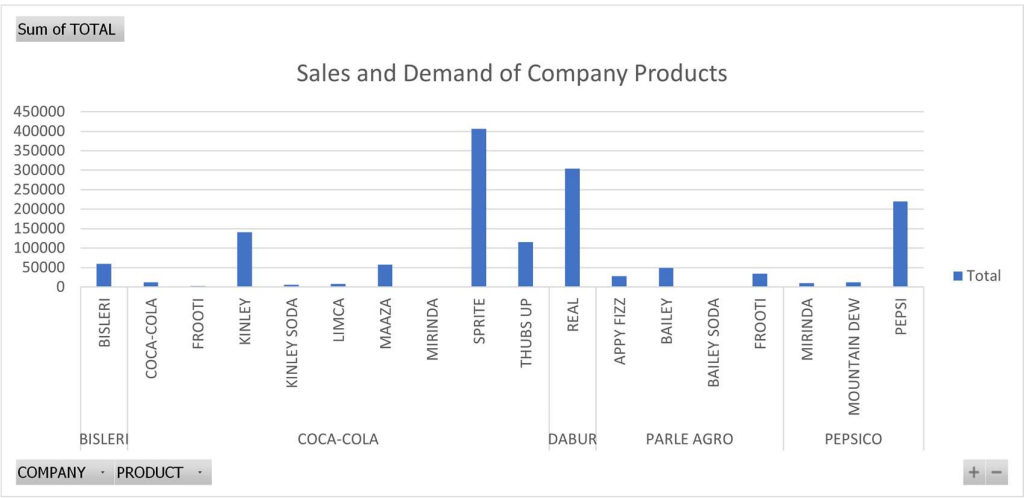
i. Monthly Demand: The monthly demand graph shows, demand of products of particular companies during the month of July, August and September. The x-axis represents the name of months and name of companies, while y-axis denotes count of company.



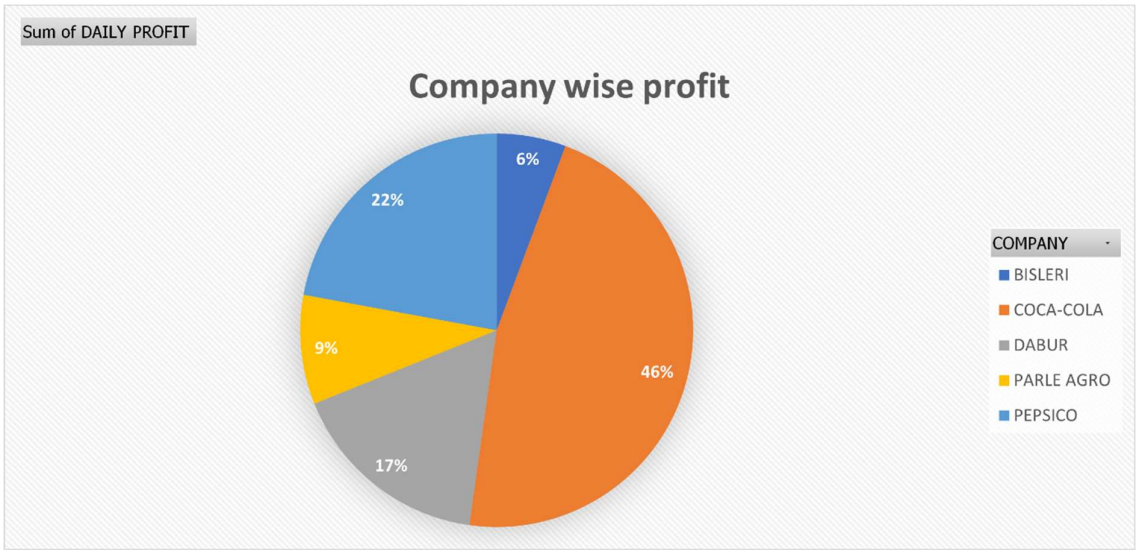
ii. Monthly Sales Trend: The monthly sales trend chart presents the sales performance over time, indicating the overall trajectory for three months. It shows a general pattern in sales. The chart demonstrate that sale was peaked in the month of July and then started declining in upcoming months i.e. Aug and Sept. This trend is influenced by many factors such as seasonal demand, events and festivals and many more. By understanding these trends allows the business to align in inventory strategies to maximize sales during slower months.



iii. Sales and Demand: This graph shows the sales and demand of particular products of a company. The x-axis represents the company and product name, whereas y-axis represents amount of total sales. This chart allows us to identify most selling product of a company to understand customer preferences and revenue distribution. By focusing on these most demanded items, the business can optimize its marketing efforts and channel resources to increase sales in slower months.



iv. Company wise profit: This chart shows the company wise profit distribution in terms of percentage. This chart shows that certain companies have higher profit distribution as compared to others. This indicates the company which demand is higher and margin is also higher. Identifying these companies helps the business to plan its prerequisites and enhance the inventory management strategies accordingly.



v. Location wise Sales: The given data represents the sales of the business across various locations. The data also includes a grand total for each location. Here x-axis represents name of locations and y-axis represents sum total for each location.

